222 Does the Commonsense Depend on Culture? Results of Simple Scripts Retrieval from WWW

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Abstract

In our research we are planning to achieve methods for automatic commonsense retrieval from the Internet resources. The first task we have chosen was to make a program to collect verb chains for nouns which would let us construct action strings conceptually similar to Schankian scripts. In this paper we introduce our preliminary experiments determining how these strings are dependent on the language and the cultural background of its user. Automatically retrieved Japanese scripts in their simplest form were used for manual comparison with Chinese, Polish and English versions of strings to investigate the level of differences and to decide if the commonsense databases for different languages should be crossed or not.

Keywords: commonsense, web-mining, scripts.

1 INTRODUCTION

1.1 Need for Commonsense Retrieval

Nowadays, amount of information increases quickly with rapid growth of the Internet. As an enormous database, it is convenient if the information which humans need is acquired from the huge source of Wide World Web. Related information can be retrieved in a matter of seconds but it is mostly knowledge which we, humans, need. Obvious knowledge called "commonsense" does not become an object of search queries as we do not need it. However the computers are told to lack this knowledge which we gather our whole lives and it is a reason that people do not treat machines as intelligent partners, especially during conversations. There are several researches of gathering commonsense as CyC [1] or OpenMind Commonsense [2] but they concentrate on manual or half-automatic processing and are done only for English language. As we assume that there is too much of such knowledge to be inputted, we try to make this process automatic by using Web-mining techniques. As we claim Japanese language to have the best predispositions for such processing thanks to its particles, we decided to use only Japanese WWW resources. Before we started we wanted to observe how usable is Japanese and how the commonsense differs depending on a language. This report is about our preliminary trials and its results.

1.2 Definition of a Script

We use the information which is retrived from WWW to create scripts automatically. Scripts[3] are the memory structures which summarize a typical phenomenon sequence into one frame, and it is defined as what expressed the standardized action which human is performing every day in the form of the chain of a phenomenon in alignment with the passage of time. For example, the restaurant scripts, usually can express that the customers "enter into restaurant", "sit down on a chair", "place an order", "eat the food", "pay money" and "leave". As it is the very beginning of our research so we treat any verb chain as a script if it has at least one noun with a particle joining a noun and a verb. Therefore the smallest unit as "watch the TV then go to bed" we also treat as a script and we made our trials on such units.

2 OUR SYSTEM

2.1 Algorithm

First, we decided to work with nouns as keywords for collecting the scripts. With them we were able to retrieve the nouns from sentences and extract the related sequences for creating dictionaries as verb dictionary, noun dictionary and other dictionaries from 1,907,086 sentences WWW corpus retrieved by Larbin robot. The noun and verb dictionaries consist of 79,460 verbs and 134,189 nouns retrieved with help of ChaSen. For creating scripts automatically, our system must search for the relationships between verbs and nouns and also between verbs. In this step, we used the verbs and nouns which had the highest occurrence, as they are used by human every day and are often used in our everyday lives, for example *television, movie, food.* As mentioned above, we used Japanese language, which has useful features like *rentaikei* where the verb suffix *te* usually joins verbs in a time sequence e.g. *gohan wo tabete neru* (to sleep after having a dinner). We can distinguish and change verb suffixes of Japanese to retrieve other, for example casual, strings for the commonsense, here we concentrated on *rentaikei* for collecting actions following one after another. This process is the first step for the more and more complex scripts creation proposed by Rzepka at al.[4]:

(dinner,bread,apple...)–*wo*–eat–*te*→
(supermarket,restaurant,garden...)–*wo*–leave–*te*→...

where *wo* is object-indicating particle. The basic processing is: a) frequency check to retrieve 3 most frequent verbs following the keyword noun, b) discovering the most frequent particle between them, c) retrieving a list of most frequent verbs occurring after the previously chosen verb, d) check if the noun-particle unit occurs with new verb-verb unit, e) if yes - the minimal script is remembered with its overall frequency.

2.2 Experiment and Results

The steps a)-c) showed that text processing on our Web corpus takes a lot of time (average 52 seconds for a query) and for longer sequences (steps d and e) even bigger version of Web corpus (9,802,434 sentences) was not enough, therefore we decided to use Google API. Because of problems with Unicode coding we gave up Linux and made next steps on Windows system (Pentium 4, CPU 3GHz, 504 MB RAM) using Perl for the text processing and Java for connecting Google servers. For example, television-*wo* watch-*te* +[sleep, cry, think...] (*terebi wo mite (neru, naku, omou...)*) takes 3 seconds (165 hits, 10 snippets retrieved) but selecting the first pair of noun and verb still must be

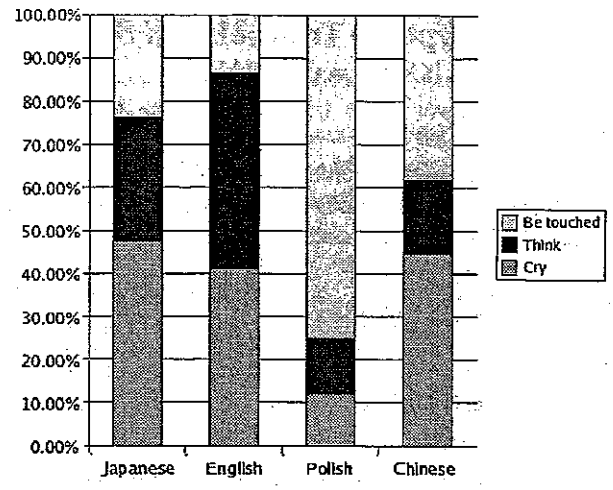


Fig. 1 Example: actions following "seeing a movie" in Japanese, English, Polish and Chinese

done with our corpus due to the Google limitations (1,000 searches per day) and takes up to one hour in very common keywords like *book*. We consider 3 most frequent actions as commonsense and in 13 picked common nouns (*subway, book, work, newspaper, baseball, bath, flu, swimming, water, bread, movie, television* and *meal*) we managed to confirm the naturalness of all retrieved trigrams based on them, though it was not always possible to collect the whole set of 3 due to the Google limits and encoding errors, for instance:
- fill with water and (boil, use, mix)
- get on the subway and (go back home, move, go)
- eating meal and (go back home, sleep, lose weight)
- play baseball (make living)
- take a bath and (*encoding error*, sleep, go out)
- read a book and (spend time, learn, think)

As the last example shows there were also cases when "te" suffix showed a nuance of a simultaneous action (*spend time reading books* not *read a book and spend time*). The data set for Google API apparently differs from the data set used by Google search engine but we found out that percentage of occurrences is similar.

3 COMPARISON WITH OTHER LANGUAGES

As mentioned in the introduction we were interested how much the different language results differ from the Japanese ones. For example, in Japanese, we discovered that 3 most frequent actions which follow *watching a TV* are:
- watching TV → sleeping: 165 times
- watching TV → laughing: 90 times
- watching TV → thinking: 74 times

We used the Google searching engine for manual frequency checks for English, Polish and Baidu searching engine for Chinese and we noticed that in most cases the proportions differ from Japanese results depending on the language, culture and different habits. Because in every language generation methods differ, as in English and Polish where it is natural to say "go by subway" rather than "get on the subway and go", or because of the Polish conjugations and genders

Table 1 Proportions of actions in different languages

act1	verb2	JPN	ENG	POL	CHN
fill	boil	305	22	3	11900
with	use	179	354	2	2
water	mix	102	40	5	3710
take	go home	120	14	7	311
the	move	30	9	10	181
subway	went to	27	170	60	883
have	go back home	288	6	0	769
a	sleep	204	13	0	269
meal	lose weight	36	3780	48	1060
watch	cry	226	1430	13	190
a	think	135	1580	14	71
movie	be touched	112	467	81	162

it is very difficult to count all the possibilities. Although we confirmed that the 4 fullest sets of trigrams had different proportions. Examined examples are shown in Tab. 1, percentage of one example is shown in Fig. 1.

4 CONCLUSIONS AND FUTURE WORK

We confirmed that Japanese language has comparatively useful structure for the commonsensical retrieval from the enormous data sets and partially confirmed that the results would differ if we make the automatic script retrieval for other languages. This time we only compared Japanese top 3 scripts with the same scripts in 3 languages but as we checked manually these top 3 results may vary. We found out that Chinese very often wash their faces and English work after watching TV. Also English results may vary depending on the origin of the users writing in English. We think our results and methods might be useful for the fields of sociology or anthropology. As this was a report about part of preparations for a bigger project, there is a lot left to be done. The next step we take is to combine simple scripts into the bigger, more elastic strings which would be useful for planning, goal discovery, etc.

References

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